

Glaucoma Overview for Deep Learning Researchers

What is Glaucoma?

Glaucoma is a group of progressive optic neuropathies characterized by retinal ganglion cell (RGC) death and visual field loss. It is the **leading cause of irreversible blindness** worldwide (~80M affected).

Types Relevant to This Project

Type	Key Feature	Relevance
Primary Open-Angle (POAG)	Most common; elevated IOP	Baseline comparison
Normal-Tension (NTG)	IOP within normal range	Common in myopes
Myopic Glaucoma	High myopia + glaucoma	This project's target

Myopic Glaucoma — The Challenge

High myopia ($\geq -6\text{D}$ spherical equivalent or axial length $\geq 26\text{mm}$) fundamentally changes the optic nerve head and retina in ways that confound standard glaucoma diagnosis and monitoring.

Why Myopia Complicates Glaucoma

- Structural changes:** Tilted discs, peripapillary atrophy (PPA), staphyloma deform the optic nerve head appearance
- RNFL thinning baseline:** Myopic eyes have thinner RNFL at baseline (mimicking glaucoma damage)
- IOP underestimation:** Thinner corneas (CCT) in myopes cause tonometry to underread IOP
- VF artifacts:** Myopic VF defects (scotomas) overlap with glaucomatous patterns
- Faster progression:** Myopic eyes tend to progress faster when glaucoma is present
- Poorer structural-functional correlation:** The standard structure-function relationship breaks down

Clinical Consequence

- Standard glaucoma grading tools (Glaucoma Hemifield Test, pattern deviation) are unreliable in myopes
- Diagnosis is often delayed — makes early prediction of **progression rate** even more critical

Key Clinical Measurements

Intraocular Pressure (IOP)

- Normal: 10–21 mmHg; in myopes, true IOP may be higher than measured
- Target for treatment; reduction slows progression

Optic Nerve Head (ONH) Parameters

- Cup-Disc Ratio (CDR):** Glaucoma → larger cup relative to disc

- **Rim Area/ISNT rule:** Inferior > Superior > Nasal > Temporal RNFL thickness (often violated in myopes)
- **Lamina Cribrosa:** Deeper insertion in myopes → more vulnerable

RNFL (Retinal Nerve Fiber Layer)

- Measured by OCT (Optical Coherence Tomography)
- Key sectors: superior, inferior, temporal, nasal
- **Global average:** single summary statistic most used in studies

Axial Length

- **Critical feature for this model** — correlates with myopia severity
- Typical: 22–25mm; High myopia: >26mm

Why MD and VFI?

Mean Deviation (MD)

- Global index of VF sensitivity vs age-matched normals
- Units: **dB** (decibels)
- Range: 0 dB (normal) to –30 dB (severe loss)
- **Progression rate:** MD change per year (dB/year)
- Clinically meaningful change: >–1 dB/year = fast progressor

Visual Field Index (VFI)

- Percentage of normal VF function remaining
- Range: 100% (normal) to 0% (end-stage)
- Less affected by cataract than MD
- **Progression rate:** VFI change per year (%/year)
- Used for estimating time-to-blindness

Why Predict Both?

- MD and VFI capture slightly different aspects of progression
- VFI is more robust to media opacities (cataract — common in elderly myopes)
- Clinical guidelines use both for treatment decisions
- Dual prediction improves model calibration

Visual Field Testing

Humphrey Field Analyzer (HFA) 24-2

- Standard for glaucoma monitoring
- Tests 54 points in central 24° visual field
- Each point measured in dB
- Strategy: SITA-Standard or SITA-Fast

Reliability Indices

Index	Acceptable	Problem
Fixation Loss	<20%	Patient not fixating
False Positive Rate	<15%	"Trigger-happy" patient
False Negative Rate	<33%	Inattentive patient

Filter unreliable tests before training — they add noise to progression estimates.

Key Formulas for This Project

OLS Progression Rate (used in preprocessing)

$$\text{rate} = \frac{\sum[(t_i - \bar{t})(VF_i - \bar{VF})]}{\sum[(t_i - \bar{t})^2]}$$

Where t_i = years from baseline, VF_i = MD or VFI at visit i

Conversion: MD to VFI (approximate)

$$VFI \approx 100 \times 10^{(MD/10)} \quad [\text{simplified; actual computation uses pointwise values}]$$

References

- Weinreb RN et al. "Primary Open-Angle Glaucoma." NEJM, 2014
- Ohno-Matsui K et al. "Myopic Glaucoma." Progress in Retinal and Eye Research, 2021
- Leung CKS. "Diagnosing Glaucoma in the Myopic Eye." Eye, 2019
- Garway-Heath DF. "Structure-Function Relationship in Glaucoma." Ophthalmology, 2002